

HARSH SAW

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SUMMARY

Full-stack software engineer with 2+ years of internship experience shipping production systems across distributed messaging, cloud infrastructure, and AI pipelines. Currently building RAG and agentic workflows at OmMuse; prior work spans real-time marketing platforms, video streaming infrastructure, and mobile apps serving thousands of users.

EDUCATION

Bachelor of Computer Information Systems, Okanagan College

Expected Dec 2026

EXPERIENCE

Software Engineer Intern
Ommuse, Seattle, US

Nov 2025 – Present

- Architecting **AI-powered semantic search** and natural language chat for enterprise studio workflows using a **RAG** pipeline (**MongoDB Vector Store**, **AWS Bedrock**), enabling producers to query projects, tracks, and collaborators conversationally.
- Built **HubSpot CRM** sync for a 12K+ user music platform in **Go** — async batch worker using **sync.Map** deduplication (30s flush, 100-user chunks) syncing live metrics across 8 API trigger points; platform manages 18K+ tracks and 67 GB of user audio.
- Engineered a cross-platform desktop uploader (**Electron**) with persistent folder-sync engine (**chokidar** + **SQLite WAL**), parallelized **S3 multipart uploads**, **OAuth 2.0 PKCE** auth flow, and a typed IPC bridge exposing 70+ **gRPC** service methods to the renderer.
- Shipped features across a polyglot monorepo (**Go gRPC/protobuf**, TypeScript/React, Python) unified by **Bazel**, owning full deployment infrastructure — **AWS** (S3, SQS, ECR), **Docker**, **Pulumi** IaC, Envoy proxy, and **GitHub Actions** CI/CD.

Full-Stack Developer Intern

Jun 2025 – Sept 2025

MoreThinks Solutions Ltd., Burnaby, BC

- Developed full-stack features using **Qwik.js** and **SurrealDB** in a monorepo, with CI/CD automation via **AWS EKS** and **API Gateway**.
- Engineered automated token generation and **vector embedding pipelines**, producing structured data consumed by ML services for recommendation.

Software Engineer Intern

Dec 2024 – Jun 2025

Bwisher Ltd, Remote

- Architected a multi-channel marketing platform using **Apache Kafka** and **BullMQ**, orchestrating personalized SMS, email, and **WhatsApp** campaigns with custom offers across **30K+ users** while feeding real-time CRM and analytics pipelines.
- Designed idempotent **NestJS** and **FastAPI** microservices for payment and credit workflows with retry and exponential backoff strategies, eliminating duplicate transactions in high-stakes financial operations.
- Owned end-to-end deployment infrastructure with **Jenkins**, **Kubernetes**, **Prometheus**, and **Grafana**; shipped features behind feature flags enabling safe tenant-by-tenant rollouts with instant kill-switch control.

Full-Stack Developer Intern

Jan 2024 – Dec 2024

Jythu Ltd, Remote

- Shipped two production **React Native** apps (LMS + admin panel) with **Node.js** backends serving **2K+ learners**; provisioned full **AWS** infrastructure with **Terraform** and **GitHub Actions** CI/CD, with multi-layer anti-piracy controls including device-bound auth.
- Built an end-to-end video streaming pipeline using **FFmpeg** HLS transcoding delivered via **S3/CloudFront**, orchestrated through **SQS**, **Lambda**, and **SNS** — reducing seek time from **30s** to **1.5s** and cutting bandwidth by approximately **70%**.

PROJECTS

Document Intelligence Platform (RAG) *FastAPI, LangChain, Qdrant, Groq, Google Gemini, AWS ECS Fargate, Docker*

[Live Demo](#) — [GitHub](#)

- Architected a multi-tenant RAG platform with session-isolated **Qdrant** indexes, enabling semantic search and conversational Q&A over PDF, DOCX, and TXT documents.
- Engineered resilient LLM orchestration using **LangChain RunnableWithFallbacks**, implementing automatic provider failover (Groq to Gemini) with streaming retrieval pipelines featuring context-aware query rewriting and hybrid document chunking.
- Deployed to **AWS ECS Fargate** with GitHub Actions CI/CD and multi-worker Uvicorn scaling; designed idempotent ingestion using **SHA-256** fingerprinting to deduplicate embeddings and reduce redundant indexing.

CodeExpo – Browser-Based IDE & Sandbox Environment *React, Node.js, WebSocket, Docker, Nginx, Zustand, xterm.js*

[Live Demo](#) — [GitHub](#)

- Engineered a full-stack browser-based IDE replicating a local dev environment — live terminal via **xterm.js** and **WebSocket**, real-time code execution with browser preview pane, and **Socket.IO** file synchronization across the workspace.
- Architected containerized sandbox environments using **Docker** with multi-stage builds and **Nginx** reverse proxy, managing isolated execution contexts and environment lifecycle per session.
- Delivered a native-feeling developer workspace with a split-pane layout (editor, terminal, live preview) powered by **Monaco Editor**; managed synchronized file tree and tab state via Zustand with **React Query** handling optimistic updates for a lag-free editing experience.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), TypeScript, Python, Go (Golang), Bash
Frontend & Mobile	React, Next.js, React Native, Qwik.js, Electron
Backend & Messaging	NestJS, FastAPI, Express, Fastify, GraphQL, gRPC/Protobuf, Apache Kafka, BullMQ, RabbitMQ
Databases	PostgreSQL, MongoDB, DynamoDB, Redis, MySQL, SurrealDB, SQLite
Cloud & DevOps	AWS, Azure, Docker, Kubernetes, Terraform, Pulumi, Bazel, Jenkins, GitHub Actions
AI & GenAI	LangChain, LangSmith, AutoGen, RAG, FAISS, Pinecone, Qdrant, AstraDB, AWS Bedrock

ACHIEVEMENTS

- Sole developer of an internal LMS migration platform at **Okanagan College**, built and deployed end-to-end and relied on daily by **1,500+ staff and instructors** for the institution-wide Moodle-to-Brightspace transition — featuring guided step-by-step migration workflows with progress tracking and complete records, real-time notifications and chat, and AI-powered templating.
- Built an end-to-end stock prediction pipeline covering **500 S&P 500 constituents** over 5 years of 15-minute bar data — **XGBoost** models trained across 26 intraday horizons on **4x NVIDIA H100 GPUs** (DRAC), with automated warm-start retraining every 15 minutes during market hours, served via a **FastAPI** backend backed by a **PostgreSQL** data warehouse and a multi-rolling feature pipeline.
- Ranked in the **top 2% of verified freelancers** on [Freelancer.com](#) with a 5-star rating, delivering 25+ production-grade projects.
- Won an inter-university college hackathon, demonstrating innovative problem-solving and technical proficiency.